

# Bromine

Br (Br I). Strongest classified transitions below 2 μm (NIST): 33 levels, 47 lines.

Br<sup>+</sup> ionisation limit 95284.80 cm<sup>-1</sup> = 11.81381 eV (104.949 nm)

Arrow label: vacuum wavelength | reduced dipole matrix element  $|U||e||J'J''|$  in ea<sub>0</sub>  
Matrix elements are measured or NIST values; \* = high-accuracy theory, ≈ = model calculation.  
Arrow width grows with the Einstein coefficient A. Dotted: forbidden (clock, M1, E2). Dashed: Rydberg excitation, n = 70.  
Level caption: energy (NIST, or a measurement of the last 20 years where one exists) and measured lifetime (where one exists).

